

Field	Details
Date	19 July 2026
Team ID	SB-AIML-2026-0042
Project Name	Emotion Detector - AI-Driven Emotion Detection & Personalized Learning Support P
Maximum Marks	3 Marks

Features List

S.No	Feature Name	Feature Description	Module / Component	Status
1	Free-Text Emotion Input	Student types study	`app/pages/home_demo.	Done
2	BiLSTM Emotion	Bidirectional LSTM	`src/models/bilstm/`	Done
3	BERT Emotion	Fine-tuned	`src/models/bert/`	Done
4	Weighted Ensemble	Combines BiLSTM (40%) (40/60) + BERT (60%) probability vectors for final		

prediction

				<code>`src/detection/ensemble.</code>	Done
5	Rule-Based	Keyword	Detects emotion-specific	<code>py`</code>	
	Boost		keywords and boosts		

corresponding probability

by 15%

6	Mixed Emotion Detection	Flags mixed emotional	<code>`src/detection/rule_based`</code> <code>.py`</code>	Done
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state when top-2 emotion

gap < 15%

`src/detection/mixed_em

otion.py`

Done

S.No	Feature Name	Feature Description	Module / Component	Status
7	Confidence Score	Shows detected emotion	`app/pages/home_demo.	Done
8	Emotion Probability Bar	Interactive Plotly bar	`src/analytics/charts.py`	Done
9	Keyword Tag Display	Shows which emotion	`app/pages/home_demo.	Done
10	Gemini-Powered	5-section personalized	`src/gemini/guidance_ge	Done
11	Emotion-Specific Prompt	Custom prompts per	`src/gemini/prompt_build	Done
12	Session CSV Logging	Full interaction metadata	`src/analytics/logger.py`	Done
13	Analytics Dashboard	4 Plotly charts: emotion	`app/pages/analytics.py`	Done
14	Model Comparison Page	Side-by-side BiLSTM vs	`app/pages/model_comp arison.py`	Done
15	Dark Glassmorphism UI	Premium dark theme with Inter font, gradient cards, frequency		

animated badges

			<code>`app/assets/style.css`</code>	Done
16	Interactive API Key Input	Users can paste Gemini		

API key directly in the UI

(no config file needed)

`app/pages/home_demo.

py`

Done

S.No	Feature Name		Feature Description	Module / Component	Status
17	Session	History	Collapsible table showing	`app/pages/home_demo.	Done
18	Streamlit	Cloud	App deployed publicly on	`streamlit_app.py`	Done
19	Retry + Backoff for	Automatic retry with	`src/gemini/client.py`		Done
20	Preprocessing Pipeline		Dual-path text cleaning	`src/preprocessing/pipeli	Done

		(aggressive for BiLSTM, ne.py`	
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Metric	Count / Value
Total Features Planned	20
Total Features Implemented	20
Core / Must-Have Features	10
Should-Have Features	6
Could-Have Features	4
Pending Features	0
Completion Rate	**100%**

Additional Features (Bonus)

Feature	Description
Gemini-Only Cloud Mode	On Streamlit Cloud, Gemini doubles as both emotion classifier + guidance generat
Negation-Preserving Stopwords	"not", "can't", "won't" etc. are preserved during cleaning to maintain sentiment

Feature	Description
Unit Test Suite	25+ pytest unit tests covering preprocessing, detection, and analytics modules
Config-Driven Architecture	All thresholds, paths, and hyperparameters centralized in <code>`config/settings.py`</code>